

Edward Huang

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EDUCATION

University of California: San Diego

BS in Computer Science (4.00 GPA) in progress

La Jolla, CA

Expected June 2028

SKILLS

Languages: Java, Python, C, C++, C#, TypeScript, JavaScript, HTML, CSS, SCSS, Lua (and learning Go)

Frontend: Angular, React, Next.js, Electron

Backend: Node.js, Express, FastAPI, OAuth, JWT, REST APIs

Technologies: SQL, SQLite, MongoDB, Redis, AWS, Google Cloud Platform, Docker, WSL, Git, Linux, Trello

EXPERIENCE

Frontend Software Designer + Developer | React, Next.js, Electron, Wavesurfer.js

Feb. 2025 - present

Engineers 4 Exploration (E4E) @ UCSD

- Built data verification interface for San Diego Zoo's bird vocalization analysis tool handling 100,000+ recordings.
- Created 25+ intuitive keybinds directly suggested by zoologists, enabling 4x increase in user efficiency.
- Renders customizable, paginated grid of spectrograms, with live audio playback and annotation.
- Added spectrogram caching system to optimize page transitions, eliminating 1000ms+ delays.
- Improved legacy frontend architecture to meet quality standards, reducing bugs and improving code modularity.

Robosub Simulations Subteam Lead | C++, ROS2, Gazebo, Linux

Feb. 2025 - present

Triton Robosub @ UCSD

- Created aquatic unmanned vehicle simulations to test path-planning algorithms and generate mock camera data.
- Implemented ROS2 nodes in C++, required source code analysis due to limited/misleading documentation.
- Wrote documentation for the subteam, detailing environment setup and development progresses.

Backend / Automation Engineer | React, Typescript, GitHub Actions

Oct. 2025 - Present

Association of Computing Machinery (ACM) @ UCSD

- Built CI/CD pipelines for Google-Sheets-to-Notion automated workflow and Discord bot using GitHub Actions.
- Automated Google Form response validation with Zod, reducing the need for human oversight.
- Implemented backend API routes for Hackathon Portal used by 100s of participants and administrators.

SDSU Big Data Hackathon: 3rd Place | Python, Flask, JavaScript

Nov. 2025

- Implemented online forum for unhoused people in San Diego, with additional chat moderator functionality.
- Helped implement map-marking feature, allowing users to mark locations of life-saving resources.

UCSD HealthLink Hackathon: 3rd Place | Python, FastAPI, Huggingface

Jun. 2025

- Led team of 4 in creating a Chrome extension that accurately identifies 85% of medical misinformation online.
- Built backend REST API endpoint that scans HTML DOM and uses RAG-based LLM to make informed corrections.

SOLO PROJECTS

Online DAW | Angular, Node.js, MongoDB, AWS S3, Redis, Auth0, Google Cloud, Vercel

July. 2025 - present

- [synthia-f.vercel.app](#) | [GitHub](#)
- Digital audio workstation and social platform, enables musicians to create and share music.
- Created 20+ visual editor features: MIDI editor, box select, drag and drop, snap-to-grid, zoom, etc.
- Implemented audio streaming + visualization + synthesization (15+ synth presets) leveraging Web Audio API.
- Leveraged MongoDB Atlas database for scalable, paginated search queries on users and projects.
- Sped up audio file queries by 60% utilizing Redis cache.
- Designed custom Angular state store by extending signals to create a deeply nested "signal tree".
 - Supports history tracking so user can undo/redo actions, following the Command pattern.
 - Created recursive factory to build type-safe signal tree from JSON using mapped types + conditional generics.
 - 50% less boilerplate than NgRx implementation.